

Lots of Authors:

Stochastic Subspace Identification and Prerequisites

Summary for supervision discussion

This note summarizes the stochastic subspace identification algorithm I intend to use to find the modes of the floe-wave system - specifically from the wave induced flexure output of the Forward Model.

1 Outline of document

This document will cover, from start to finish, all the prerequisite and actual contents of the Stochastic Subspace Identification algorithm. It begins with deterministic models (where the input forcing is known), and progresses to impulse response only identification and finally to fully unknown inputs. This is a very control theoretic themed exercise, so I will to the best of my ability explain all notation and terms. Nonetheless, in the interest of myself not spending more than a day writing this up, it will not be as extensive as it could be. Thus please send me a message if anything is unclear.

Much content here will be familiar to you as it essentially boils down to applied linear algebra/ODEs.

2 Basic control theory

From a control systems perspective, system identification is the process of uncovering an "untouched" system's dynamics. These are governed by the physics involved, such as the equation of motion for a spring-mass-damper or the thermodynamics involved determining a room's temperature. Knowledge of how the physical system behaves allows control theorists to design controllers (external elements which change the system's dynamics) to alter the system's response. If the system is fully known, a controller can be designed to ensure the system will respond exactly how you want for a given input.

The major system characteristics of interest are:

1. Accuracy: Given a reference input (setpoint) which you want your system to track, how well does it do this (l_2 norm between the reference and measured response).
2. Settling time τ : The amount of time until the system reaches 63% of the setpoint.
3. Damping ratio ζ : Characterises the behaviour of the system in the transient phase; i.e. will it overshoot the setpoint before tracking or not. More complicated than this but for now this is enough.

I will not explain any more on these specifications, as they are not essential for now. The point is, we can design controllers which allow us to satisfy these if we have full knowledge of the system.

Classical control theory uses full knowledge of the input forcing to derive something called a **transfer function**. You can think of this as a transformation $T : \mathbb{R} \rightarrow \mathbb{C}$ mapping how the output y responds given some input u . It is defined as the Laplace transform of these two quantities, set up as output over input.

For instance, take the spring-mass-damper (u = forcing, y = displacement, k = spring constant, c = damping constant, m = mass):

$$m \frac{d^2 x(t)}{dt^2} + c \frac{dx(t)}{dt} + kx(t) = u(t)$$

Taking the Laplace transform of both sides and solving for the transfer function yields:

$$H(s) = \frac{Y(s)}{U(s)} = \frac{1}{ms^2 + cs + k}$$

The poles of this function tell us exactly how it behaves. Note $s = \sigma + j\omega$. Note that this is very similar to the transfer function for floating bodies, namely the relative amplitude operator:

$$\mathbf{RAO}(\omega, \beta) = \frac{\mathbf{X}(\omega, \beta)}{\zeta_a} = [-\omega^2 (\mathbf{M} + \mathbf{A}(\omega)) + i\omega \mathbf{B}(\omega) + \mathbf{C}]^{-1} \mathbf{F}_{ex}(\omega, \beta)$$

Naturally this is a bit more complex, as it I have included all 6 degrees of freedom, but the idea is the same. Also note that here I took the Fourier not Laplace transform. The forcing involved will not be wide-banded so this makes sense!

To solve for the system's response, one just needs to find all the coefficients involved, multiply the transfer function by the input forcing, and inverse Laplace/Fourier transform the result.

3 Balanced Model Reduction

Moving away from this transfer function notion is necessary for higher degree of freedom systems, specifically from a computational perspective. State-spaces are used to efficiently capture all of the relevant dynamics. Let $\mathbf{x}$ be a vector of system states, i.e. it contains the minimum variables required to fully describe the system. With $\mathbf{y}$ and $\mathbf{u}$ being the output and input respectively, the state-space realisation of a system is governed by the following matrix system of differential equations:

$$\dot{\mathbf{x}}(t) = \underline{\underline{\mathbf{A}}}\mathbf{x}(t) + \underline{\underline{\mathbf{B}}}\mathbf{u}(t)$$

$$\mathbf{y}(t) = \underline{\underline{\mathbf{C}}}\mathbf{x}(t) + \underline{\underline{\mathbf{D}}}\mathbf{u}(t)$$

It is annoying to keep the double underlines, just note all capital A, B, C and D refer to matrices please.

What **balanced model reduction** does is assuming we have the state matrices A, B and C (assumes D = 0 i.e. no feedforward terms) and that as of now $\mathbf{x}$ is super high dimensional, we want to define a new system of considerably lower dimension:

$$\dot{\tilde{\mathbf{x}}} = \underline{\underline{\tilde{\mathbf{A}}}}\tilde{\mathbf{x}} + \underline{\underline{\tilde{\mathbf{B}}}}\mathbf{u}$$

$$\mathbf{y} = \underline{\underline{\tilde{\mathbf{C}}}}\tilde{\mathbf{x}}$$

Moreover, assume there exists some map Ψ from $\mathbf{x}$ to $\tilde{\mathbf{x}}$

$$\mathbf{x} = \underline{\underline{\Psi}}\tilde{\mathbf{x}}$$

3.1 Grammians

The way this map is constructed is through the manipulation of the **controllability** and **observability** Grammians. These are measures of how much control effort it takes to push a system into a specific state, and how prominent a state is in measurements respectively. These aspects are captured in their eigenvalues. If a state aligns with a large eigenvalue in $\underline{\underline{\mathbf{W}_c}}$, it takes very little energy to move the system there, thus it is highly controllable, and if a state aligns with a large eigenvalue in $\underline{\underline{\mathbf{W}_o}}$, a tiny movement in that state creates a MASSIVE spike in the sensors, thus it is highly observable.

The rotation matrix $\underline{\underline{\Psi}}$ ensures that the state space is rotated such that these two Grammians become equal and diagonal, whose diagonal elements are the Hankel singular values sorted largest to smallest.

Letting T be the transformation from $\mathbf{x}$ to $\mathbf{z}$, we see it's effect on the Grammians as:

$$\begin{aligned}\hat{W}_c &= \int_0^\infty e^{\hat{A}\tau} \hat{B} \hat{B}^* e^{\hat{A}^*\tau} d\tau = T^{-1} \int_0^\infty e^{A\tau} \underbrace{TT^{-1}}_I BB^* \underbrace{T^{-*}T^*}_I e^{A^*\tau} d\tau T^{-*} = T^{-1} W_c T^{-*} \\ \hat{W}_o &= \int_0^\infty e^{\hat{A}^*\tau} \hat{C}^* \hat{C} e^{\hat{A}\tau} d\tau = T^* \int_0^\infty e^{A^*\tau} \underbrace{T^{-*}T^*}_I C^* C \underbrace{T^{-1}T}_I e^{A\tau} d\tau T = T^* W_o T\end{aligned}$$

Thus we can see it is a simple left and right multiplication by the transformation matrix to recover the new rotated diagonal and equal Grammians. There is a lot more to this, but the point is we can reduce the order of our system by considering the most controllable and observable variable set from the full set $\mathbf{x}$. Using the SVD makes this even smaller when you truncate low value singular values.

The issue right now is that Grammians are of $O(n^3)$ to compute. **Balanced proper orthogonal decomposition** solves this issue by considering the adjoint system (note I have moved to discrete time)

Direct system:

$$\mathbf{x}_{k+1} = \underline{\underline{\mathbf{A}}}\mathbf{x}_k + \underline{\underline{\mathbf{B}}}\mathbf{u}_k = \underline{\underline{\mathbf{C}}}\mathbf{x}_k$$

Adjoint system:

$$\mathbf{x}_{k+1} = \underline{\underline{\mathbf{A}}}^*\mathbf{x}_k + \underline{\underline{\mathbf{C}}}^*\mathbf{y}_k$$

Now instead of calculating the Grammians directly, we instead can use the impulse response. This is done via spiking the input $\mathbf{u}$ in the direct system and the output $\mathbf{y}$ in the adjoint.

This gives rise to 2 new matrices, the controllability matrices; the **controllability matrix** $\mathcal{C}$ and **observability matrix** $\mathcal{O}$:

$$\mathcal{C} = [B \quad AB \quad A^2B \quad \dots \quad A^mB]$$

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^m \end{bmatrix}$$

Note that these are precisely the stored effects of an impulse response. Try for yourself, plug in $\mathbf{u}_k = \mathbf{I}$ for $k = 1$ and 0 else, and that will derive the controllability matrix. Same if you spike y in the adjoint space.

The Grammians can be shown to reduce to:

$$\begin{aligned} W_c &\approx \mathcal{C}\mathcal{C}^* \\ W_o &\approx \mathcal{O}^*\mathcal{O} \end{aligned}$$

Which holds for small Δt and $m \gg 1$. And as mentioned before, there is a way of representing the reduced order model in terms of these Grammians and a transformation matrix. It is not important to know how to calculate these as we will not be using this method.

4 Eigensystem Realisation Algorithm

This is an algorithm used extensively in vibration analysis of large structures. It is an extremely linked to the BPOD. Here we assume we can measure $\mathbf{u}$ and $\mathbf{y}$ but not $\mathbf{x}$ directly. The goal is similar, we will attempt to learn a rank r approximation of the system dynamics, except here we do not know the system matrices!

Spiking the input with $\mathbf{u} = [1, 0, 0, \dots, 0]$ yields:

k	u	x	y
0	I	0	D
1	0	B	CB
2	0	AB	CAB
3	0	A^2B	CA^2B
$\vdots$	$\vdots$	$\vdots$	$\vdots$
k	0	$A^{k-1}B$	$CA^{k-1}B$

and note that:

$\mathbf{y}$	DATA
D	y_0
CB	y_1
CAB	y_2
CA^2B	y_3
$\vdots$	$\vdots$
$CA^{k-1}B$	y_k

The first step in the ERA is to stack the outputs into a Hankel matrix (equal off-diagonal elements):

$$H = \begin{bmatrix} y_1 & y_2 & y_3 & \dots \\ y_2 & y_3 & y_4 & \dots \\ y_3 & y_4 & y_5 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

and one timestep onwards:

$$H' = \begin{bmatrix} y_2 & y_3 & y_4 & \dots \\ y_3 & y_4 & y_5 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Here lies the connection between the ERA and BPOD:

$$H = \overbrace{\begin{bmatrix} \textcolor{violet}{CB} & \textcolor{violet}{CAB} & y_3 & \dots \\ y_1 & y_2 & y_4 & \dots \\ y_2 & y_3 & y_5 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}}^{\mathcal{O}\mathcal{C} \text{ (from BPOD)}} = \mathcal{O}\mathcal{C}$$

and

$$H' = \begin{bmatrix} \textcolor{violet}{CAB} & \textcolor{violet}{CA^2B} & y_4 & \dots \\ \textcolor{violet}{y_2} & \textcolor{violet}{y_3} & y_5 & \dots \\ y_3 & y_4 & y_5 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} = \mathcal{O}AC$$

Then just take the SVD and truncate (will only keep $\tilde{r}$ elements):

$$H = \begin{bmatrix} \textcolor{violet}{CB} & \textcolor{violet}{CAB} & y_3 & \dots \\ \textcolor{violet}{y_1} & \textcolor{violet}{y_2} & y_4 & \dots \\ y_2 & y_3 & y_4 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} = U\Sigma V^T = [\tilde{U} \quad U_t] \begin{bmatrix} \tilde{\Sigma} & 0 \\ 0 & \Sigma_t \end{bmatrix} \begin{bmatrix} \tilde{V}^T \\ V_t^T \end{bmatrix}$$

Thus as before, it is easiest to choose our truncation index r based off the singular values (i.e. the diagonal elements of Σ).

Through some hectic linear algebra you get the following:

$$\begin{aligned} \tilde{A} &= \tilde{\Sigma}^{-1/2} \tilde{U}^T H' \tilde{V} \tilde{\Sigma}^{-1/2} \\ \tilde{B} &= \tilde{\Sigma}^{1/2} \tilde{V}^T \begin{bmatrix} \mathbb{I}_p & 0 \\ 0 & 0 \end{bmatrix} \\ \tilde{C} &= \begin{bmatrix} \mathbb{I}_q & 0 \\ 0 & 0 \end{bmatrix} \tilde{U} \tilde{\Sigma}^{1/2} \end{aligned}$$

And thus, purely from impulse response data and no clue of the system dynamics, you can reconstruct a low order approximation of your system.

5 Stochastic Subspace Identification

This is a purely output-only identification technique, i.e. no system matrices, no $\mathbf{u}$, no impulse response. The states themselves are assumed to be a stationary, zero-mean stochastic process. The state space representation thereof is given by:

$$\begin{aligned} \mathbf{x}_{k+1} &= \mathbf{A}\mathbf{x}_k + \mathbf{w}_k \\ \mathbf{y}_k &= \mathbf{C}\mathbf{x}_k + \mathbf{v}_k \end{aligned}$$

5.1 Notation and Properties of Stochastic LTI systems

Where $\mathbf{w}_k$ and $\mathbf{v}_k$ are unmeasurable stochastic noise elements due to disturbances/modeling inaccuracies and measurement errors respectively. They are assumed to white, zero mean and have covariance matrices:

$$E \left[\begin{pmatrix} \mathbf{w}_p \\ \mathbf{v}_p \end{pmatrix} \begin{pmatrix} \mathbf{w}_q^T & \mathbf{v}_q^T \end{pmatrix} \right] = \begin{pmatrix} \mathbf{Q} & \mathbf{S} \\ \mathbf{S}^T & \mathbf{R} \end{pmatrix} \delta_{pq}$$

Below are other properties and definitions required:

$$\begin{aligned} E[\mathbf{w}_k] &= \mathbf{0}, & E[\mathbf{v}_k] &= \mathbf{0} \\ E[\mathbf{x}_k] &= \mathbf{0}, & E[\mathbf{x}_k \mathbf{x}_k^T] &= \Sigma \\ E[\mathbf{x}_k \mathbf{w}_k^T] &= \mathbf{0}, & E[\mathbf{x}_k \mathbf{v}_k^T] &= \mathbf{0} \end{aligned}$$

And the output covariance (Λ) and next-state output covariance (G) matrices are defined by ($\mathbf{E}$ is the expectation operator):

$$\begin{aligned} \Lambda_i &\equiv E[\mathbf{y}_{k+i} \mathbf{y}_k^T] \in \mathbb{R}^{l \times l} \\ \mathbf{G} &\equiv E[\mathbf{x}_{k+1} \mathbf{y}_k^T] \in \mathbb{R}^{n \times l} \end{aligned}$$

Yielding the following covariance relations/properties:

$$\begin{aligned} \Sigma &= \mathbf{A}\Sigma\mathbf{A}^T + \mathbf{Q} \\ \Lambda_0 &= \mathbf{C}\Sigma\mathbf{C}^T + \mathbf{R} \\ \mathbf{G} &= \mathbf{A}\Sigma\mathbf{C}^T + \mathbf{S} \\ \Lambda_i &= \mathbf{C}\mathbf{A}^{i-1}\mathbf{G} \end{aligned}$$

Pay special attention to the last formula:

$$\mathbf{\Lambda}_i = \mathbf{C}\mathbf{A}^{i-1}\mathbf{G}$$

Notice that it implies that the output covariances can be considered as impulse responses of the deterministic linear time-invariant system $\mathbf{A}$, $\mathbf{G}$, $\mathbf{C}$, $\mathbf{\Lambda}_0$.

Assume we have l sensors (or at least channels, for a single IMU $l = 6$). Designate r of them to be **reference** sensors. Thus you may write the output and covariance matrices as:

$$\begin{aligned}\mathbf{y}_k &\equiv \begin{pmatrix} \mathbf{y}_k^{\text{ref}} \\ \mathbf{y}_k^{\sim\text{ref}} \end{pmatrix}, \quad \mathbf{y}_k^{\text{ref}} = \mathbf{L}\mathbf{y}_k, \quad \mathbf{L} \equiv [\mathbf{I}_r, \mathbf{0}] \\ \mathbf{\Lambda}_i^{\text{ref}} &\equiv E[\mathbf{y}_{k+i}(\mathbf{y}_k^{\text{ref}})^T] = \mathbf{\Lambda}_i\mathbf{L}^T \in \mathbb{R}^{l \times r} \\ \mathbf{G}^{\text{ref}} &\equiv E[\mathbf{x}_{k+1}(\mathbf{y}_k^{\text{ref}})^T] = \mathbf{G}\mathbf{L}^T \in \mathbb{R}^{n \times r}\end{aligned}$$

Now that important equation becomes:

$$\mathbf{\Lambda}_i^{\text{ref}} = \mathbf{C}\mathbf{A}^{i-1}\mathbf{G}^{\text{ref}}$$

Define the Hankel partition matrix as:

$$H \equiv \frac{1}{\sqrt{j}} \begin{pmatrix} y_0^{\text{ref}} & y_1^{\text{ref}} & \cdots & y_{j-1}^{\text{ref}} \\ y_1^{\text{ref}} & y_2^{\text{ref}} & \cdots & y_j^{\text{ref}} \\ \vdots & \vdots & \ddots & \vdots \\ y_{i-1}^{\text{ref}} & y_i^{\text{ref}} & \cdots & y_{i+j-2}^{\text{ref}} \\ y_i & y_{i+1} & \cdots & y_{i+j-1} \\ y_i & y_{i+2} & \cdots & y_{i+j} \\ \vdots & \vdots & \ddots & \vdots \\ y_{2i-1} & y_{2i} & \cdots & y_{2i+j-2} \end{pmatrix} \equiv \begin{pmatrix} Y_{0|i-1}^{\text{ref}} \\ Y_{i|2i-1} \end{pmatrix} \equiv \begin{pmatrix} Y_p^{\text{ref}} \\ Y_f \end{pmatrix} \begin{matrix} \updownarrow r \\ \updownarrow l \end{matrix} \begin{matrix} \text{"past"} \\ \text{"future"} \end{matrix} \in \mathbb{R}^{(r+l)i \times j}$$

Another partition is garnered by adding one block row to the past references and omitting the first block row of future outputs:

$$H = \begin{pmatrix} Y_{0|i}^{\text{ref}} \\ Y_{i|i}^{\sim\text{ref}} \\ Y_{i+1|2i-1} \end{pmatrix} = \begin{pmatrix} Y_p^{\text{ref}+} \\ Y_{i|i}^{\sim\text{ref}} \\ Y_f^- \end{pmatrix} \begin{matrix} \updownarrow r(i+1) \\ \updownarrow l-r \\ \updownarrow l(i-1) \end{matrix}$$

The extended observability matrix has a similar definition as well as the stochastic extended controllability matrix:

$$\mathbf{O}_i \equiv \begin{pmatrix} \mathbf{C} \\ \mathbf{C}\mathbf{A} \\ \mathbf{C}\mathbf{A}^2 \\ \vdots \\ \mathbf{C}\mathbf{A}^{i-1} \end{pmatrix} \in \mathbb{R}^{li \times n}, \quad \mathbf{C}_i^{\text{ref}} \equiv \begin{pmatrix} \mathbf{A}^{i-1}\mathbf{G}^{\text{ref}} & \mathbf{A}^{i-2}\mathbf{G}^{\text{ref}} & \cdots & \mathbf{A}\mathbf{G}^{\text{ref}} & \mathbf{G}^{\text{ref}} \end{pmatrix} \in \mathbb{R}^{n \times ri}$$

5.2 Reference-Based Covariance-Driven SSI

5.2.1 Covariance-Driven SSI

This is a modified version of the SSI-COV, which required covariance calculations between all elements of the output, now only between a finite set of references. Gather the output covariances into a Toeplitz matrix (equal diagonals):

$$T_{1|i}^{\text{ref}} \equiv \begin{pmatrix} \Lambda_i^{\text{ref}} & \Lambda_{i-1}^{\text{ref}} & \cdots & \Lambda_1^{\text{ref}} \\ \Lambda_{i+1}^{\text{ref}} & \Lambda_i^{\text{ref}} & \cdots & \Lambda_2^{\text{ref}} \\ \vdots & \vdots & \ddots & \vdots \\ \Lambda_{2i-1}^{\text{ref}} & \Lambda_{2i-2}^{\text{ref}} & \cdots & \Lambda_i^{\text{ref}} \end{pmatrix} \in \mathbb{R}^{li \times ri}$$

Assuming ergodicity yields:

$$T_{1|i}^{\text{ref}} = Y_f Y_p^{\text{ref}T}$$

Using the important equation let's us write:

$$T_{1|i}^{\text{ref}} = \begin{pmatrix} C \\ CA \\ \vdots \\ CA^{i-1} \end{pmatrix} (A^{i-1}G^{\text{ref}} \quad A^{i-2}G^{\text{ref}} \quad \dots \quad AG^{\text{ref}} \quad G^{\text{ref}}) = O_i C_i^{\text{ref}}$$

(sorry these are matrices, too lazy to bold now. Succinctly:

$$\mathbf{T}_{1|i}^{\text{ref}} = \mathbf{Y}_f (\mathbf{Y}_p^{\text{ref}})^T = \mathbf{O}_i \mathbf{C}_i^{\text{ref}}$$

One could then take the SVD of this matrix:

$$T_{1|i}^{\text{ref}} = USV^T = (U_1 \quad U_2) \begin{pmatrix} S_1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} V_1^T \\ V_2^T \end{pmatrix} = U_1 S_1 V_1^T$$

where $U \in \mathbb{R}^{li \times li}$ and $V \in \mathbb{R}^{ri \times ri}$ are orthonormal matrices ($U^T U = U U^T = I_{li}$, $V^T V = V V^T = I_{ri}$), and $S \in \mathbb{R}^{li \times ri}$ contains singular values in descending order. Truncating to rank n yields the reduced system representation.

Now, balanced factorization is acquired by equating $T_{1|i}^{\text{ref}} = O_i C_i^{\text{ref}}$ with the truncated SVD product $U_1 S_1 V_1^T$, the extended observability and stochastic controllability matrices are realized in a balanced coordinate system:

$$\begin{aligned} O_i &= U_1 S_1^{1/2} \\ C_i^{\text{ref}} &= S_1^{1/2} V_1^T \end{aligned}$$

We can recover the state matrix $\underline{\underline{A}}$ from the shifted Toeplitz matrix:

$$T_{2|i+1}^{\text{ref}} = O_i A C_i^{\text{ref}}$$

and the observability and controllability matrices:

$$A = O_i^\dagger T_{2|i+1}^{\text{ref}} C_i^{\text{ref}\dagger} = S_1^{-1/2} U_1^T T_{2|i+1}^{\text{ref}} V_1 S_1^{-1/2}$$

5.2.2 Reference-Driven SSI

Now instead of all past outputs we will only use the past reference outputs, reducing computational time.

The Kalman filter has the goal of producing an optimal prediction for the state vector $\mathbf{x}_{k+1}$. This is accomplished by using observations of the outputs up to time k , along with the available system matrices and known noise covariances.

With initial conditions $\hat{x}_0 = 0$ and $P_0 \equiv E[\hat{x}_0 \hat{x}_0^T] = 0$, the state estimate, gain matrix, and error covariance update recursively as:

$$\begin{aligned} \hat{\mathbf{x}}_{k+1} &= \mathbf{A} \hat{\mathbf{x}}_k + \mathbf{K}_k (\mathbf{y}_k - \mathbf{C} \hat{\mathbf{x}}_k) \\ \mathbf{K}_k &= (\mathbf{G} - \mathbf{A} \mathbf{P}_k \mathbf{C}^T) (\mathbf{\Lambda}_0 - \mathbf{C} \mathbf{P}_k \mathbf{C}^T)^{-1} \\ \mathbf{P}_{k+1} &= \mathbf{A} \mathbf{P}_k \mathbf{A}^T + (\mathbf{G} - \mathbf{A} \mathbf{P}_k \mathbf{C}^T) (\mathbf{\Lambda}_0 - \mathbf{C} \mathbf{P}_k \mathbf{C}^T)^{-1} (\mathbf{G} - \mathbf{A} \mathbf{P}_k \mathbf{C}^T)^T \end{aligned}$$

The Kalman filter states are captured in their own matrix:

$$\hat{\mathbf{X}}_i \equiv (\hat{\mathbf{x}}_i \quad \hat{\mathbf{x}}_{i+1} \quad \dots \quad \hat{\mathbf{x}}_{i+j-1}) \in \mathbb{R}^{n \times j}$$

Taking the QR factorisation of the block Hankel matrix of past references and future outputs:

$$H \equiv \begin{pmatrix} Y_p^{\text{ref}} \\ Y_f \end{pmatrix} = R Q^T$$

where $Q \in \mathbb{R}^{j \times j}$ is orthonormal ($Q^T Q = Q Q^T = I_j$) and $R \in \mathbb{R}^{(r+l)i \times j}$ is lower triangular.

The orthogonal projection of the row space of future outputs Y_f onto the row space of past reference outputs Y_p^{ref} is defined as:

$$\mathcal{P}_i^{\text{ref}} \equiv \mathbf{Y}_f / \mathbf{Y}_p^{\text{ref}} \equiv \mathbf{Y}_f (\mathbf{Y}_p^{\text{ref}})^T (\mathbf{Y}_p^{\text{ref}} (\mathbf{Y}_p^{\text{ref}})^T)^{\dagger} \mathbf{Y}_p^{\text{ref}} = \begin{pmatrix} \mathbf{R}_{21} \\ \mathbf{R}_{31} \\ \mathbf{R}_{41} \end{pmatrix} \mathbf{Q}_1^T \in \mathbb{R}^{li \times j}$$

The main theorem of SSI ensures that the projection matrix $\mathcal{P}_i^{\text{ref}}$ factorizes into the extended observability matrix O_i and the Kalman filter state sequence $\hat{X}_i$

$$\mathcal{P}_i^{\text{ref}} = \begin{pmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{i-1} \end{pmatrix} (\hat{x}_i \quad \hat{x}_{i+1} \quad \dots \quad \hat{x}_{i+j-1}) \equiv O_i \hat{X}_i$$

The order n system can be reduced via SVD:

$$\mathcal{P}_i^{\text{ref}} = U_1 S_1 V_1^T$$

where $U_1 \in \mathbb{R}^{li \times n}$, $S_1 \in \mathbb{R}^{n \times n}$, and $V_1 \in \mathbb{R}^{j \times n}$. The extended observability matrix and Kalman state estimates are recovered as:

$$O_i = U_1 S_1^{1/2} \\ \hat{X}_i = O_i^\dagger \mathcal{P}_i^{\text{ref}}$$

The system matrices are derived now. Define another projection:

$$\mathcal{P}_{i-1}^{\text{ref}} \equiv Y_f^- / Y_p^{\text{ref}+} = \begin{pmatrix} R_{41} & R_{42} \end{pmatrix} \begin{pmatrix} Q_1^T \\ Q_2^T \end{pmatrix} \in \mathbb{R}^{l(i-1) \times j}$$

which can be shown to equal:

$$\mathcal{P}_{i-1}^{\text{ref}} = O_{i-1} \hat{X}_{i+1}$$

where $O_{i-1} \in \mathbb{R}^{l(i-1) \times n}$ is obtained by deleting the last l rows of O_i . The time-shifted Kalman state sequence is recovered via:

$$\hat{X}_{i+1} = O_{i-1}^\dagger \mathcal{P}_{i-1}^{\text{ref}}$$

5.2.3 Deriving system matrices for reference SSI

Given the recovered Kalman state sequences $\hat{X}_i$ and $\hat{X}_{i+1}$, the system matrices A and C are identified by solving the overdetermined set of state and observation equations:

$$\begin{pmatrix} \hat{X}_{i+1} \\ Y_{i|i} \end{pmatrix} = \begin{pmatrix} A \\ C \end{pmatrix} \hat{X}_i + \begin{pmatrix} \rho_w \\ \rho_v \end{pmatrix}$$

where $Y_{i|i} \in \mathbb{R}^{l \times j}$ is the single-block output Hankel matrix. In terms of the RQ -decomposition factors:

$$Y_{i|i} = \begin{pmatrix} R_{21} & R_{22} & 0 \\ R_{31} & R_{32} & R_{33} \end{pmatrix} \begin{pmatrix} Q_1^T \\ Q_2^T \\ Q_3^T \end{pmatrix} \in \mathbb{R}^{l \times j}$$

Since the residuals $(\rho_w^T \quad \rho_v^T)^T$ are uncorrelated with the state estimates $\hat{X}_i$, A and C are solved in a least-squares sense using the Moore-Penrose pseudo-inverse:

$$\begin{pmatrix} A \\ C \end{pmatrix} = \begin{pmatrix} \hat{X}_{i+1} \\ Y_{i|i} \end{pmatrix} \hat{X}_i^\dagger$$

The noise covariance matrices Q, R, S are computed directly as the sample covariances of the residuals $\begin{pmatrix} \rho_w \\ \rho_v \end{pmatrix}$. To convert (A, C, Q, R, S) into the stochastic state-output parameter set (A, G, C, Λ_0) , the state covariance matrix Σ is solved via the discrete Lyapunov equation:

$$\Sigma = A \Sigma A^T + Q$$

Once Σ is known, G and Λ_0 are determined by:

$$\Lambda_0 = C \Sigma C^T + R \\ G = A \Sigma C^T + S$$

6 Postprocessing

6.1 Modal Analysis

A mode is characterised by an eigenfrequency, a damping ratio, a mode shape and a modal scaling factor. Note that in output-only modal analysis, this last parameter cannot be estimated.

The above method will allow us to identify the state matrix $\underline{\mathbf{A}}$, where the dynamic behaviour of the system is entirely described by its eigenvalues:

$$\mathbf{A} = \Psi \Lambda \Psi^{-1}$$

where $\Lambda = \text{diag}(\lambda_q) \in \mathbb{C}^{n \times n}$ ($q = 1, \dots, n$) is a diagonal matrix containing the complex discrete-time eigenvalues, and $\Psi \in \mathbb{C}^{n \times n}$ contains the corresponding eigenvectors as columns.

The continuous-time state matrix A_c and its relationship to the discrete state matrix A via sampling time Δt :

$$\begin{aligned} A_c &= \Psi_c \Lambda_c \Psi_c^{-1} \\ A &= \exp(A_c \Delta t) \\ \Psi_c &= \Psi, \quad \lambda_{c_q} = \frac{\ln(\lambda_q)}{\Delta t} \end{aligned}$$

The continuous-time eigenvalues occur in complex conjugate pairs and yield the modal damping ratio ξ_q and eigenfrequency ω_q (rad/s) of mode q :

$$\lambda_{c_q}, \lambda_{c_q}^* = -\xi_q \omega_q \pm j \omega_q \sqrt{1 - \xi_q^2}$$

The physical mode shapes $\Phi \in \mathbb{C}^{l \times n}$ observed at the output sensor locations are mapped via the observation matrix C :

$$\Phi = C \Psi$$

6.2 Model order selection: the stabilization diagram

Everything above assumed the model order n is simply known, via a clean gap between retained and discarded singular values. In practice, with real (noisy) data this gap is rarely visible, and choosing n this way is unreliable. The standard practical fix, used throughout the literature this method comes from, is the **stabilization diagram**: the identification above is repeated at many increasing model orders $n = 1, 2, \dots, n_{\max}$, and the resulting poles are tracked from one order to the next. A pole is declared *stable* if it reappears at a similar frequency, damping ratio, and mode shape as n increases; typical thresholds (Peeters and De Roeck, 1999) are 1% change in frequency, 5% change in damping ratio, and a mode shape consistency above 99% (see MAC below). Genuine physical modes stabilize quickly and persist across a wide range of orders; noise, and any pole not corresponding to real structural dynamics, tends to wander or appear only intermittently.

6.2.1 The Modal Assurance Criterion (MAC)

Comparing mode shapes across orders (or against a reference shape) needs a similarity measure between two complex vectors that is insensitive to arbitrary scaling, since mode shapes from output-only identification are only known up to an unknown scale factor. The Modal Assurance Criterion does this:

$$\text{MAC}(\Phi_a, \Phi_b) = \frac{|\Phi_a^H \Phi_b|^2}{(\Phi_a^H \Phi_a)(\Phi_b^H \Phi_b)} \in [0, 1], \quad (1)$$

where $\Phi_a, \Phi_b \in \mathbb{C}^l$ are two candidate mode shape vectors (columns of $\Phi = C \Psi$ from the previous section) and $(\cdot)^H$ denotes the conjugate transpose. $\text{MAC} = 1$ indicates the two shapes are perfectly consistent (up to scale and phase); $\text{MAC} = 0$ indicates orthogonal, unrelated shapes. In the stabilization diagram, Φ_a, Φ_b are the same physical mode's identified shape at two consecutive orders n and $n + 1$; a pole is only counted as having a stable mode shape if its MAC value exceeds the chosen threshold.

6.2.2 Relevance here

This is the concrete mechanism for separating the three pole classes expected in this project: genuine floe flexural modes, radiation-memory poles, and input-color artifacts from the non-white JONSWAP forcing. All three are, in principle, genuine poles of *some* linear system fitted to the data, so no single-order SVD truncation can distinguish them. Tracking stability across order, together with prior knowledge of the approximate JONSWAP peak frequency, is what allows a spurious forcing-shaped pole to be identified and discarded rather than mistaken for a structural mode.

6.2.3 Spectral analysis

The power spectral density matrix of a stochastic system can be shown to be:

$$\mathbf{S}_{yy}(z) = [\mathbf{C}(z\mathbf{I}_n - \mathbf{A})^{-1}\mathbf{G} + \mathbf{A}_0 + \mathbf{G}^T(z^{-1}\mathbf{I}_n - \mathbf{A}^T)^{-1}\mathbf{C}^T]_{z=e^{j\omega\Delta t}}$$

6.2.4 Prediction errors and modal response

By solving a Riccati equation, the stochastic state-space model converts into forward innovation form:

$$\begin{aligned} z_{k+1} &= Az_k + Ke_k \\ y_k &= Cz_k + e_k \end{aligned}$$

where $K \in \mathbb{R}^{n \times l}$ is the steady-state Kalman gain and $e_k \in \mathbb{R}^{l \times 1}$ is the innovation vector with covariance $E[e_p e_q^T] = R_e \delta_{pq}$. Transforming into modal coordinates via $z_{m_k} = \Psi^{-1}z_k$, $K_m = \Psi^{-1}K$, and $\Phi = C\Psi$ yields:

$$\begin{aligned} z_{m_{k+1}} &= \Lambda z_{m_k} + K_m e_k \\ y_k &= \Phi z_{m_k} + e_k \end{aligned}$$

Eliminating innovations $e_k = y_k - \hat{y}_k$ gives the predictor system driven by measured outputs y_k :

$$\begin{aligned} z_{m_{k+1}} &= (\Lambda - K_m \Phi) z_{m_k} + K_m y_k \\ \hat{y}_k &= \Phi z_{m_k} \end{aligned}$$

where $\hat{y}_k$ is the one-step-ahead output estimate. The contribution of individual mode q to output response, and the overall measured output decomposition, are expressed as:

$$\begin{aligned} \hat{y}_k^{(q)} &= \Phi_q(z_{m_k})_q \\ y_k &= \sum_{q=1}^n \Phi_q(z_{m_k})_q + e_k \end{aligned}$$